HDOC DAY 76

1. Create an enumeration for user roles (ADMIN, USER, GUEST) and display messages based on role.
    a. Input: GUEST
    b. Output: Welcome Guest
2. Print all enum names and integer values of 12 colors using a loop.
    a. Input: No Input
    b. Output: RED=0, BLUE=1, GREEN=2

1. C Program to Create Enumeration for User Roles

Program:

```
#include <stdio.h>
#include <string.h>

enum Role {
   ADMIN, USER, GUEST
};

int main() {
   char role[20];

   printf("Enter role: ");
   scanf("%s", role);

   enum Role r;

   if (strcmp(role, "ADMIN") == 0)
      r = ADMIN;
```

```
    else if (strcmp(role, "USER") == 0)
        r = USER;
    else if (strcmp(role, "GUEST") == 0)
        r = GUEST;
    else {
        printf("Invalid Role");
        return 0;
    }

    switch (r) {
        case ADMIN:
            printf("Welcome Admin");
            break;

        case USER:
            printf("Welcome User");
            break;

        case GUEST:
            printf("Welcome Guest");
            break;
    }

    return 0;
}
```

Algorithm:

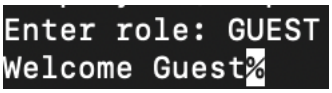
1. Start the program.
2. Define an enumeration named Role with ADMIN, USER and GUEST.
3. Read the role from the user.
4. Compare the input with the available roles.
5. Assign the corresponding enum value.
6. Use a switch statement to display the welcome message.
7. Stop the program.

Test Case Table:

Input	Expected Output

ADMIN	Welcome Admin
USER	Welcome User
GUEST	Welcome Guest
MANAGER	Invalid Role

Output:



2. C Program to Print 12 Colors Using Enumeration and Loop

Program:

```
#include <stdio.h>

enum Colors {
   RED, BLUE, GREEN, YELLOW,
   ORANGE, PURPLE, BLACK, WHITE,
   PINK, BROWN, GREY, VIOLET
};

int main() {
   enum Colors c;

   char *names[] = {
      "RED", "BLUE", "GREEN", "YELLOW",
      "ORANGE", "PURPLE", "BLACK", "WHITE",
      "PINK", "BROWN", "GREY", "VIOLET"
   };

   for (c = RED; c <= VIOLET; c++) {
```

```
        printf("%s=%d\n", names[c], c);
    }

    return 0;
}
```

Algorithm:

1. Start the program.
2. Declare an enumeration named Colors containing 12 colors.
3. Create a string array containing all color names.
4. Declare an enum variable c.
5. Use a for loop from RED to VIOLET.
6. Print each color name and its integer value.
7. Stop the program.

Test Case Table:

Test Case	Input	Expected Output
1	No Input	RED=0
2	No Input	BLUE=1
3	No Input	GREEN=2
4	No Input	YELLOW=3
5	No Input	ORANGE=4
6	No Input	PURPLE=5
7	No Input	BLACK=6
8	No Input	WHITE=7

9	No Input	PINK=8
10	No Input	BROWN=9
11	No Input	GREY=10
12	No Input	VIOLET=11

Output:

```
RED=0
BLUE=1
GREEN=2
YELLOW=3
ORANGE=4
PURPLE=5
BLACK=6
WHITE=7
PINK=8
BROWN=9
GREY=10
VIOLET=11
```

Execution

Compilation:

gcc program.c -o program

Run:

./program

Result: Both programs successfully demonstrate the use of enumerations in C, including role-based message display and printing enum names with their integer values using a loop.